

5.2.25

EE25BTECH11022 - sankeerthan

Question: Solve the following system of linear equations $ax + by = c$, $bx + ay = 1 + c$
Solution: Given

$$ax + by = c \quad (1)$$

$$bx + ay = 1 + c \quad (2)$$

The matrix equation for a line is defined as

$$\mathbf{n}^\top \mathbf{x} = c \quad (3)$$

where $\mathbf{n}$ is the coefficient matrix and $\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix}$

$$\begin{pmatrix} a & b \end{pmatrix} \mathbf{x} = c \quad (4)$$

$$\begin{pmatrix} b & a \end{pmatrix} \mathbf{x} = 1 + c \quad (5)$$

As a matrix equation

$$\begin{pmatrix} a & b \\ b & a \end{pmatrix} \mathbf{x} = \begin{pmatrix} c \\ 1 + c \end{pmatrix} \quad (6)$$

$$\begin{pmatrix} a & b \\ b & a \end{pmatrix}^\top \begin{pmatrix} a & b \\ b & a \end{pmatrix} \mathbf{x} = \begin{pmatrix} a & b \\ b & a \end{pmatrix}^\top \begin{pmatrix} c \\ 1 + c \end{pmatrix} \quad (7)$$

$$\begin{pmatrix} a & b \\ b & a \end{pmatrix} \begin{pmatrix} a & b \\ b & a \end{pmatrix} \mathbf{x} = \begin{pmatrix} a & b \\ b & a \end{pmatrix} \begin{pmatrix} c \\ 1 + c \end{pmatrix} \quad (8)$$

$$\begin{pmatrix} a^2 + b^2 & 2ab \\ 2ab & a^2 + b^2 \end{pmatrix} \mathbf{x} = \begin{pmatrix} b + (a + b)c \\ a + (a + b)c \end{pmatrix} \quad (9)$$

$$\mathbf{x} = \frac{1}{(a^2 - b^2)^2} \begin{pmatrix} a^2 + b^2 & -2ab \\ -2ab & a^2 + b^2 \end{pmatrix} \begin{pmatrix} b + (a + b)c \\ a + (a + b)c \end{pmatrix} \quad (10)$$

$$\mathbf{x} = \frac{1}{(a^2 - b^2)^2} \begin{pmatrix} (a^2 + b^2)(b + (a + b)c) - 2ab(a + (a + b)c) \\ -2ab(b + (a + b)c) + (a^2 + b^2)(a + (a + b)c) \end{pmatrix} \quad (11)$$

$$\mathbf{x} = \frac{1}{(a^2 - b^2)^2} \begin{pmatrix} b(a^2 + b^2 - 2a^2) + ((a + b)c)(a^2 + b^2 - 2ab) \\ ((a + b)c)(a^2 + b^2 - 2ab) + a(a^2 + b^2 - 2b^2) \end{pmatrix} \quad (12)$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \frac{1}{(a^2 - b^2)^2} \begin{pmatrix} ((a + b)c)(a - b)^2 - b(a^2 - b^2) \\ ((a + b)c)(a - b)^2 + a(a^2 - b^2) \end{pmatrix} \quad (13)$$

$$(14)$$

Hence $x = \frac{c}{a+b} + \frac{b}{b^2-a^2}$, $y = \frac{c}{a+b} + \frac{a}{a^2-b^2}$ is the solution for given system of linear equations

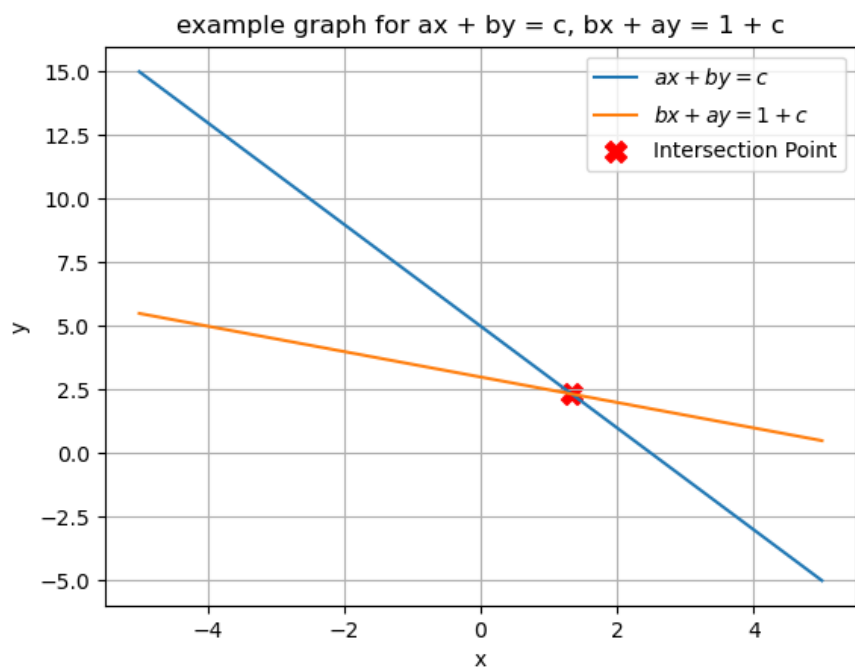


Fig. 0.1